Variational inequalities and their finite element solutions

a genuine introduction

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outline

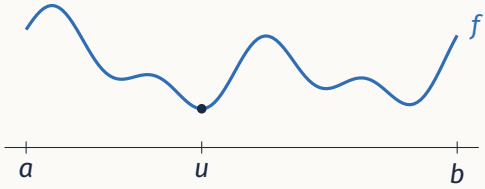
1. finite-dimensional examples ← *geometry of VIs*
2. infinite-dimensional examples ← *some applications of VIs*
3. finite element solutions ← *basics, and differences from PDE case*

- I welcome questions at any time!
- they are a *good* use of time

a calculus I problem

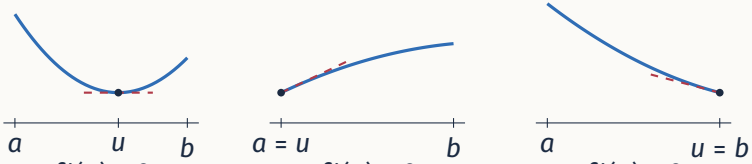
- how do you minimize a smooth $f(x)$ over a closed interval $[a, b]$?
- what we tell our students:

solve $f'(x) = 0$,
but test the end-points



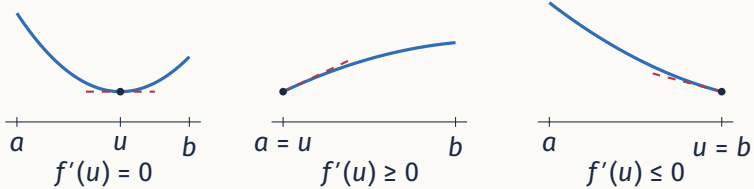
a calculus I problem

- how do you minimize a smooth $f(x)$ over a closed interval $[a, b]$?
- 3 cases for a local minimizer:



a calculus I problem

- how do you minimize a smooth $f(x)$ over a closed interval $[a, b]$?
- 3 cases for a local minimizer:



- all are captured by a single variational inequality (VI):

$$u \text{ locally minimizes } f \implies \boxed{f'(u)(x-u) \geq 0} \quad \text{for all } x \in [a, b]$$

- note 1st-order term in Taylor expansion: $f(x) = f(u) + f'(u)(x-u) + \ldots$

  viewpoint: traveling infinitesimally from u into $[a, b]$, f ~~increases~~ does not decrease

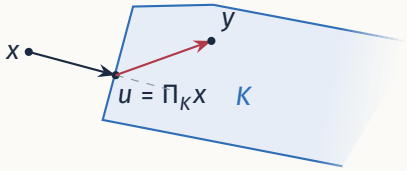
projection onto a closed, convex set

- suppose H is a real Hilbert space (e.g. $\mathbb{R}^n$)
- let $K \subset H$ be nonempty, closed, and convex

Definition

For $x \in H$, define the *projection of x onto K*:

$$u \leftarrow \min_{y \in K} \|y - x\|.$$



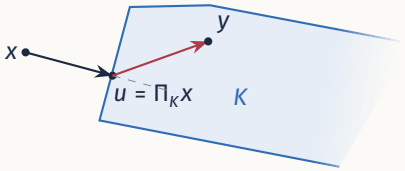
- a parallelogram law & completeness argument shows that a unique minimizer exists; this is a good definition
- notation: $u = \Pi_K x$
- observation: $\Pi_K : H \to K$ is Lipschitz continuous

projection onto a closed, convex set

Definition

For $K \subset H$ nonempty, closed, and convex, in a Hilbert space H, define the *projection of* $x \in H$ *onto* K:

$$\Pi_K x \leftarrow \min_{y \in K} \|y - x\|.$$



- question: can we characterize $\Pi_K x$ using inner products?
- answer: $u = \Pi_K x$ if and only if $u \in K$ and

$$\boxed{\langle u - x, y - u \rangle \geq 0} \quad \text{for all } y \in K$$

idea: angle between $u - x$ and $y - u$ is at most 90°

minimize f over a convex set $K \subset \mathbb{R}^n$

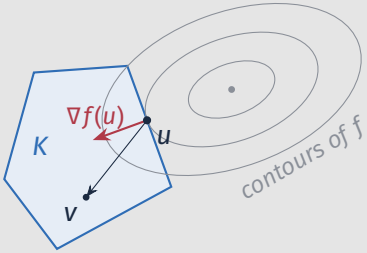
Theorem

Suppose:

- $K \subset \mathbb{R}^n$ *is convex, and*
- $f : K \to \mathbb{R}$ *is in* C^1.

If $u \in K$ *is a local minimizer then*

$$\boxed{\langle \nabla f(u), v - u \rangle \geq 0} \qquad \text{for all } v \in K.$$



proof: Note that $(1-t)u + tv = u + t(v-u) \in K$ for $0 \leq t \leq 1$.
But $t = 0$ locally minimizes $g(t) = f(u + t(v-u))$ on $[0, 1]$.
By the calculus I problem, $0 \leq g'(0) = \langle \nabla f(u), v - u \rangle$. □

idea: $\nabla f(u) \neq 0$ is possible, but f does not decrease as we move from u into K

minimize convex f over a convex set $K \subset \mathbb{R}^n$

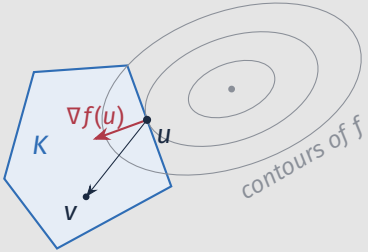
Theorem (narrowed to a characterization)

Suppose:

- $K \subset \mathbb{R}^n$ *is convex, and*
- $f : K \to \mathbb{R}$ *is in* C^1 *and convex.*

Then $u \in K$ *is a minimizer if and only if*

$$\boxed{\langle \nabla f(u), v - u \rangle \geq 0} \qquad \text{for all } v \in K.$$



- the boxed variational inequality is one way to state the KKT condition
- u exists if additionally $K \neq \emptyset$ and
  - K is compact, or
  - K is closed and f is *coercive*: $f \to \infty$ as $\|v\| \to \infty$ for $v \in K$
- u is unique if additionally f strictly convex

systems of nonlinear equations

- variational inequalities are merely convenient notation in optimization?

systems of nonlinear equations

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- but they need not come from optimization
- to put us in the right frame of mind, recall this ...

folk theorem

(n equations in n unknowns) = good

systems of nonlinear equations

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given continuous $F : \mathbb{R}^n \to \mathbb{R}^n$, there exists x so that $F(x) = 0$

systems of nonlinear equations

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~~given continuous $F : \mathbb{R}^n \to \mathbb{R}^n$, there exists x so that $F(x) = 0$~~
given continuous $F : K \to \mathbb{R}^n$, for $K \subset \mathbb{R}^n$ compact and convex,
there exists $x \in K$ so that $F(x) = 0$

systems of nonlinear equations

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~~given continuous $F : K \to \mathbb{R}^n$, for $K \subset \mathbb{R}^n$ compact and convex,~~
~~there exists $x \in K$ so that $F(x) = 0$~~

- no general existence theorem for "$F(x) = 0$," even on very nice input sets
  advice to numerical users: Newton's method with line search is powerful, but ...

constrained nonlinear equations

- minimal problem change: solve a variational inequality instead of "$F(u) = 0$"

Theorem (Hartman & Stampacchia, 1966)

If $K \subset \mathbb{R}^n$ is nonempty, compact, and convex, and if $F : K \to \mathbb{R}^n$ is continuous, then there exists $u \in K$ so that

$$\boxed{\langle F(u), v - u\rangle \geq 0} \qquad \text{for all } v \in K.$$

proof: The map $v \mapsto \Pi_K\big(v - F(v)\big)$ is continuous from K to K. By Brouwer it has a fixed point:

$$u = \Pi_K\big(u - F(u)\big).$$

Let $z = u - F(u)$ so $u = \Pi_K z$. By the projection theorem, for all $v \in K$,

$$0 \leq \langle u - z, v - u\rangle = \langle F(u), v - u\rangle . \qquad \square$$

constrained nonlinear equations

Theorem (Hartman & Stampacchia, 1964)

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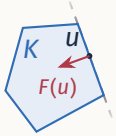
Furthermore, if u is in the interior of K then $F(u) = 0$.

proof of "Furthermore": Take $v = u + \epsilon w \in K$ for $w \in \mathbb{R}^n$ and $\epsilon > 0$ small enough. □

- exactly 2 cases:



we found u solving $F(u) = 0$



we found a support plane for K

- F here plays the role of ∇f in optimization, but its Jacobian need not be symmetric

uniqueness?

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- what about uniqueness? sufficient hypothesis on F?

uniqueness?

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Furthermore, if u is in the interior of K then $F(u) = 0$.

- what about uniqueness? sufficient hypothesis on F?

- suppose for $u_1, u_2 \in K$: $\langle F(u_i), v - u_i \rangle \geq 0$ for all $v \in K$
- substitute $v = u_2$ and then $v = u_1$, then add:

$$\langle F(u_1) - F(u_2), u_1 - u_2 \rangle \leq 0 \qquad \leftarrow \text{for uniqueness, block this when } u_1 \neq u_2$$

a well-posed problem

Definition

$F : K \to \mathbb{R}^n$ is monotone if

$$\langle F(v) - F(w), v - w \rangle \geq 0$$

for all $v, w \in K$, and strictly monotone if equality implies $v = w$.

Corollary

If additionally F is strictly monotone then $u \in K$ is unique.

- but is this hypothesis reasonable?

a well-posed problem

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Corollary

If additionally F is strictly monotone then $u \in K$ is unique.

- but is this hypothesis reasonable?
- yes, it is the same as strict convexity of the objective function

Proposition (optimization case)

Suppose $f : \mathbb{R}^n \to \mathbb{R}$ is C^1. Then f is strictly convex if and only if $F = \nabla f$ is strictly monotone.

variational inequalities

Definitions (made official)

Given a normed vector space X, a closed and convex admissible subset $K \subset X$, and $F : K \to X'$ continuous, but generally nonlinear, find $u \in K$ so that

$$\boxed{\langle F(u), v - u \rangle \geq 0} \qquad \text{for all } v \in K.$$

This is the variational inequality problem for the data K and F.

example problem	typical admissible set	variational inequality
calculus minimization	$[a, b] \subset \mathbb{R}$	$f'(u)(v - u) \geq 0 \quad \forall v \in [a, b]$
projection of x onto K	$K \subset H$ closed, convex	$\langle u - x, v - u \rangle \geq 0 \quad \forall v \in K$
minimize f over K	$K \subset \mathbb{R}^n$ compact, convex	$\langle \nabla f(u), v - u \rangle \geq 0 \quad \forall v \in K$
constrain nonlinear eqns to K	$K \subset \mathbb{R}^n$ compact, convex	$\langle F(u), v - u \rangle \geq 0 \quad \forall v \in K$

perspective: VI generalizes system $F(u) = 0$ to convex input sets, enabling existence

outline

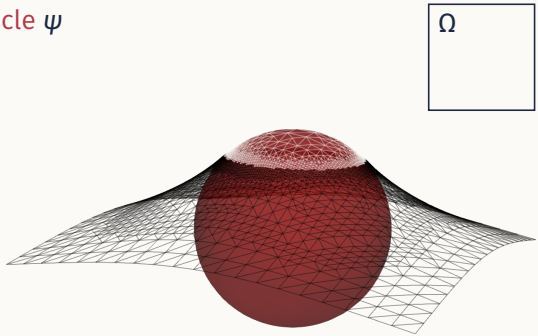
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classical obstacle problem (naive form)

- bounded domain $\Omega \subset \mathbb{R}^d$ with Lipschitz boundary
- consider an elastic membrane, with vertical displacement u over Ω
  - suppose small displacements u (we will use the Laplacian ∇^2)
- $u = g$ on $\partial\Omega$ (fixed boundary condition)
- subject to a vertical force f
- constrained to be on or above an obstacle ψ



ψ = (upper hemisphere)

classical obstacle problem (naive form)

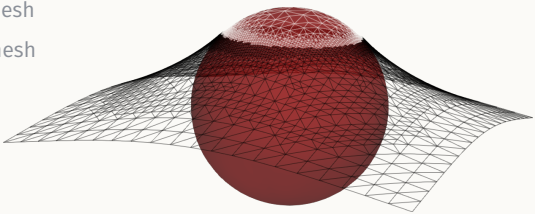
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- then:

$$-\nabla^2 u = f \text{ where } u > \psi \quad \leftarrow \text{black mesh}$$
$$u = \psi \text{ otherwise} \quad \leftarrow \text{white mesh}$$

- but *where* will $u = \psi$ hold?



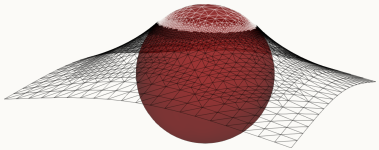
ψ = (upper hemisphere)

classical obstacle problem (strong form and subsets)

- previous problem statement too naive
- the obstacle problem can be stated as a complementarity problem:

$$u - \psi \geq 0$$
$$-\nabla^2 u - f \geq 0$$
$$(u - \psi)(-\nabla^2 u - f) = 0$$

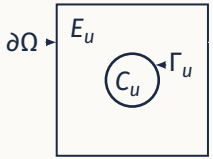
  - each holds a.e. over Ω
  - analogous to a PDE strong form



- the solution u defines subsets of Ω:

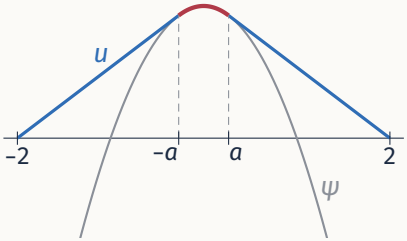
inactive set	$E_u = \{x : u(x) > \psi(x)\}$
active (contact) set	$C_u = \Omega \setminus E_u$
free boundary	$\Gamma_u = \Omega \cap \partial E_u$

  - Poisson equation $-\nabla^2 u = f$ applies in E_u
  - both Dirichlet & Neumann conditions hold along Γ_u (which compensates for unknown location of Γ_u?)



example in 1d

- suppose $\Omega = (-2, 2) \subset \mathbb{R}^1$, $f = 0$, $g = 0$
- choose obstacle $\psi(x) = 1 - x^2$
- define $K = \{v : v(\pm 2) = 0 \text{ and } v \geq \psi\}$
  - ignoring regularity for now
- seek $u \in K$ solving $-u'' = 0$ if $u > \psi$, and $u = \psi$ otherwise



result:

- $a = 2 - \sqrt{3} \approx 0.2679$ locates the free boundary
- both Dirichlet and Neumann conditions hold there: $u(a) = \psi(a)$ & $u'(a) = \psi'(a)$
- $u(x) = \begin{cases} 1 - x^2, & 0 \leq |x| \leq a, \\ 2a(2 - |x|), & a \leq |x| \leq 2, \end{cases}$
- $u \in C^1 \setminus C^2$, despite C^∞ data and Laplacian operator

classical obstacle problem (weak forms)

- for theoretical and finite-element purposes, we want to rewrite in weak form
  this step is not as obvious to applied users as the equality-constrained case?
- let $K = \{v \in X = H^1(\Omega) : v \geq \psi \text{ and } v|_{\partial\Omega} = g\}$, a closed, convex subset (not subspace)

weak forms

the classical obstacle problem seeks $u \in K$ solving

- the minimization problem:

$$u \leftarrow \min_{v \in K} \int_\Omega \frac{1}{2}|\nabla v|^2 - fv$$

- the variational inequality (VI):

$$\int_\Omega \nabla u \cdot \nabla(v - u) \geq \int_\Omega f(v - u) \quad \forall v \in K$$

Proposition

- *if $u \in K$ is regular enough to solve the complementarity problem (strong form) then it solves these weak forms*
- *the weak forms are equivalent, and well-posed*

theory interruption

Definition

A bilinear form $a(\cdot,\cdot)$ is coercive on a Hilbert space H if there is $\alpha > 0$ so that

$$a(v,v) \geq \alpha \|v\|^2 \quad \text{for all } v \in H.$$

- by the Poincaré inequality the bilinear form in the obstacle problem, $a(u,v) = \int_\Omega \nabla u \cdot \nabla v$ is coercive on $H_0^1(\Omega)$
- coercive $\implies$ strictly monotone

Theorem (Stampacchia 1964; generalizes Lax–Milgram to $K \subset H$)

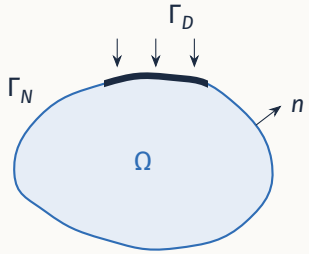
Suppose $a(\cdot,\cdot)$ is bilinear, continuous, and coercive on a Hilbert space H. Suppose $K \subset H$ is nonempty, closed, and convex. Then for $\ell \in H'$ there exists a unique $u \in K$ with

$$a(u, v-u) \geq \langle \ell, v-u \rangle \quad \forall v \in K.$$

elastic contact problem (Signorini, 1933?)

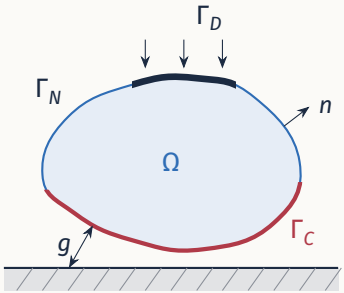
- consider a linear elastic solid on $\Omega \subset \mathbb{R}^d$ ($d = 2, 3$)
- $u : \Omega \to \mathbb{R}^d$ is *displacement* from ref. config.
- $E(u) = \frac{1}{2}\left(\nabla u + \nabla u^T\right)$ is *strain tensor*
- $\sigma(u) = \lambda\,(\operatorname{tr} E(u))\, I + 2\mu\, E(u)$ is *stress tensor*
  (Lamé constants λ, μ; see Ciarlet (2022) for elasticity)
- partition $\partial\Omega$:

  on Γ_D, a given displacement $u = u_D$,
  on Γ_N, zero traction



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  (Lamé constants λ, μ; see Ciarlet (2022) for elasticity)
- partition $\partial\Omega$:
  - on Γ_D, a given displacement $u = u_D$,
  - on Γ_N, zero traction, and
  - on Γ_C, allow frictionless contact
- gap function g is given on Γ_C
- equations of equilibrium:



(g = normal distance from ref. config. to obstacle)

$$-\nabla \cdot \sigma(u) = f \quad \text{in } \Omega$$ stress balance (f is body force)

$$u = u_D \quad \text{on } \Gamma_D$$ given displacement

$$u \cdot n \le g \quad \text{on } \Gamma_C$$ non-penetration constraint

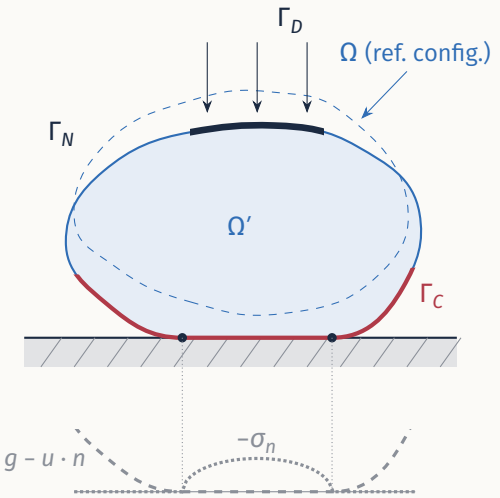
Signorini contact problem (complementarity)

- the solution displacement u warps the ref. config. Ω to new state $\Omega' = (I + u)(\Omega)$ →
- $\sigma_n = n \cdot \sigma(u)\, n$ denotes the *normal stress*
- along Γ_C:

$$
\begin{aligned}
g - u \cdot n &\geq 0 && \text{non-penetration} \\
-\sigma_n &\geq 0 && \text{inward normal stress} \\
(g - u \cdot n)\, \sigma_n &= 0 && \text{complementarity}
\end{aligned}
$$

- the free boundary is a subset of Γ_C
- frictionless contact: tangential $\sigma_t = 0$ on Γ_C

see textbook Kikuchi & Oden (1988) for contact



Signorini contact problem (VI form)

- define convex and closed admissible set:

$$K = \{v \in H^1(\Omega; \mathbb{R}^d) : v = u_D \text{ on } \Gamma_D \text{ and } v \cdot n \le g \text{ on } \Gamma_C\}$$

- VI problem is to find $u \in K$ so that

$$\int_\Omega \sigma(u) : E(v - u) \ge \int_\Omega f \cdot (v - u) \qquad \text{for all } v \in K$$

- equivalent to minimizing the elastic energy over admissible displacements:

$$u \leftarrow \min_{v \in K} \int_\Omega \tfrac{1}{2}\sigma(v) : E(v) - f \cdot v$$

Theorem (Fichera 1963)

The Signorini problem is well-posed.

- key idea: $a(v, w) = \int_\Omega \sigma(v) : E(w)$ is coercive on $H^1_0(\Omega; \mathbb{R}^d)$ by Korn's inequality

contact in nonlinear hyperelasticity (multiple solutions)

- linear elasticity is inadequate for large deformations
- consider nonlinear hyperelastic material:

$$u \leftarrow \min_{v \in K} J(v) = \int_\Omega W(I + \nabla v) - f \cdot v$$

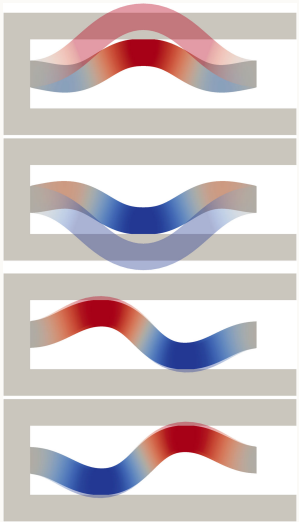
where $W(Z)$ is the *strain energy density*

  - *compressible neo-Hookean* model gives W formula
  - linearizing W at $Z = I$ recovers previous model

- a minimizer solves the VI

$$\langle J'(u), v - u \rangle \geq 0 \qquad \text{for all } v \in K$$

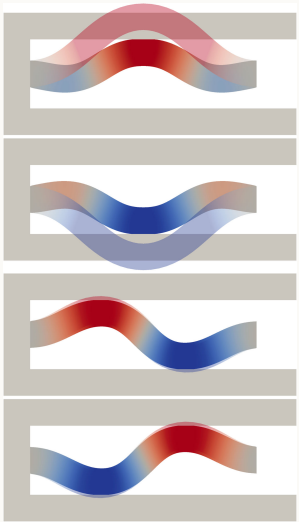
- but W is not convex in neo-Hookean model (Ciarlet 2022), so multiple minimizers are possible



Farrell, Croci & Surowiec (2020)

contact in nonlinear hyperelasticity (multiple solutions)

- Farrell, Croci & Surowiec (2020) considered a neo-Hookean beam
- constrained by top and bottom (box) obstacles
- clamped at left, compressed axially from right
- they found 7 solutions to a finite element approximation of this VI problem
  - 3 solutions never touch the obstacles (not shown)
  - 4 are in contact (right), with pale unconstrained versions
- from a new deflation technique
  - and using Firedrake, of course



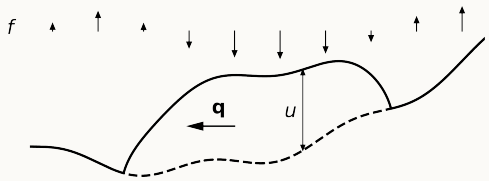
Farrell, Croci & Surowiec (2020)

fluid layer in a climate

- there are many applications of VIs, but here is my obsession ...
- consider a thin layer of moving incompressible fluid, on a substrate, and in a climate:

$$u_t + \nabla \cdot \mathbf{q} = f$$

  - u is the layer thickness
  - $\mathbf{q}(x, u, \nabla u)$ is the vertically-summed flux, e.g. gravity-driven or wind/ocean-driven
  - $f(t, x)$ is an (abstract) signed climate, a source term; it adds or removes mass



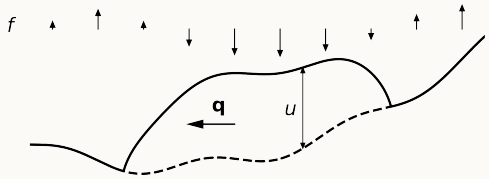
- examples: glaciers and ice sheets, sea ice, evaporating lakes, rain on a windshield
- the source f is the "surface mass balance", the difference between mass added *(e.g. precipitation)* and removed *(e.g. evaporation, melt-and-runoff)*

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- examples: glaciers and ice sheets, sea ice, evaporating lakes, rain on a windshield
- the source f is the "surface mass balance", the difference between mass added *(e.g. precipitation)* and removed *(e.g. evaporation, melt-and-runoff)*
- but $u \geq 0$, because it is a *thickness*, right?

fluid layer in a climate: implicit time step

- consider backward Euler time steps (parabolic form open?)
- for $X = W^{1,p}(\Omega)$ let $K = \{v \in X : v \geq 0\}$ be the admissible thicknesses
- given previous (time t_{n-1}) thickness $u_{n-1} \in K$
- VI problem is to find $u_n \in K$ at t_n so that, for all $v \in K$,

$$\int_\Omega \frac{u_n - u_{n-1}}{\Delta t}(v - u_n) - \mathbf{q}_n \cdot \nabla(v - u_n) \geq \int_\Omega f_n (v - u_n)$$

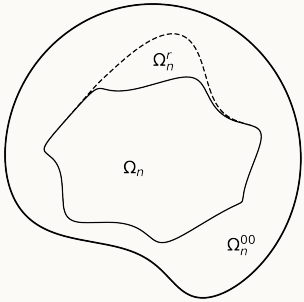
- $\mathbf{q}_n = \mathbf{q}(x, u_n, \nabla u_n)$ denotes the unknown flux
- well-posed for many flux formulas (B. 2021)
  - advective, diffusive, nonlinear, ...
- generally not an optimization
- note: $\Delta t \to \infty$ limit is steady state



fluid layer in a climate: implicit time step

VI model for updated thickness $u_n \in K = \{v \in X : v \geq 0\}$:

$$\int_\Omega \frac{u_n - u_{n-1}}{\Delta t}(v - u_n) - \mathbf{q}_n \cdot \nabla(v - u_n) \geq \int_\Omega f_n (v - u_n)$$



- the solution decomposes Ω into disjoint sets:

$\Omega_n = \{u_n > 0\}$	fluid-covered area
$\Omega_n^r = \{u_n = 0 \text{ and } u_{n-1} > 0\}$	retreat area
$\Omega_n^{00} = \{u_n = 0 \text{ and } u_{n-1} = 0\}$	remaining uncovered

- note that $u_{n-1} + \Delta t\, f_n \leq 0$ on $\Omega_n^r \cup \Omega_n^{00}$

observation (B. 2021)

The mass lost in the retreat area Ω_n^r arose from an unknowable split between flow and climate. This is a barrier to exact mass conservation in numerical simulations.

summary: ∞-dimensional VI examples

problem	unknown	constraint	minimization?
classical obstacle	displacement $u \in H^1(\Omega)$	$u \geq \psi$ on Ω	yes (convex)
Signorini contact	displacement $u \in H^1(\Omega; \mathbb{R}^d)$	$u \cdot n \leq g$ on Γ_C	yes (convex)
hyperelastic contact	"	"	yes (not convex)
fluid layer in climate	thickness $u \in W^{1,p}(\Omega)$	$u \geq 0$ on Ω	no

in each:

- one solves a PDE on an inactive subset which is not known in advance
- the VI problem does not refer, a priori, to the active set or free boundary

there are many more applications of VIs: pricing of options, traffic flow networks, game theory, electrical circuits, phase transformations, ...

outline

1. finite-dimensional examples ← *geometry of VIs*

2. infinite-dimensional examples ← *some applications of VIs*

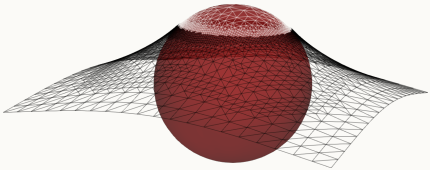
3. finite element solutions ← *basics, and differences from PDE case*

finite elements (FE) for VIs: only some basics

- the final slides in this talk:
  - are minimal and elementary regarding FE, and they
  - focus on the contrast between PDEs and VIs
- recall the continuum classical obstacle problem:

$$K = \{v \in H^1(\Omega) : v \geq \psi \text{ and } v|_{\partial\Omega} = g\}$$

$$\int_\Omega \nabla u \cdot \nabla(v - u) \geq \int_\Omega f(v - u) \quad \forall v \in K$$



- regularity for this problem:

Proposition (Kinderlehrer & Stampacchia, 1980)

For reasonable Ω, if $\psi \in H^2$ and $f \in L^2$ then $u \in H^1_g(\Omega) \cap H^2(\Omega)$. However, even if data are arbitrarily smooth, generally $u \notin C^2$ and $u \notin H^3$.

FE for VIs: the basics

- the FE problem is a finite-dimensional VI over $X_h \subset X = H^1(\Omega)$, for $u_h \in K_h$:

$$K_h = \{v_h \in X_h : v_h \geq \psi_h\}$$
$$\int_\Omega \nabla u_h \cdot \nabla(v_h - u_h) \geq \int_\Omega f(v_h - u_h) \quad \forall v_h \in K_h$$

  - setting fixed boundary condition $g = 0$ for simplicity, and assuming $\int f v_h$ is exact
  - assume for now that $\psi_h = I_h \psi$: the discrete obstacle is the interpolant of ψ
- set $X_h = P_1$ (continuous piecewise-linear)
  - over meshes $\mathcal{T}_h$ of triangles or tetrahedra

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  - over meshes $\mathcal{T}_h$ of triangles or tetrahedra
- regarding this wimpy choice:

the low-order gauntlet for VIs (Ciarlet 2002, p 289)

[Because of the low regularity in the obstacle problem], *finite elements "of low degree" are sufficient for all practical purposes, a fact amply confirmed by numerical evidence.*

FE for VIs: Firedrake for classical obstacle problem

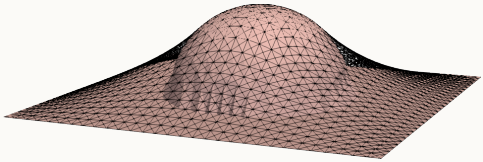
- having chosen P_1, the Firedrake (firedrakeproject.org) code is easy!

```
from firedrake import *

mesh = RectangleMesh(30, 30, 2, 2, -2, -2, diagonal="crossed")
x, y = SpatialCoordinate(mesh)
r2 = x ** 2 + y ** 2
psi = conditional(r2 < 1, sqrt(1 - r2), 0)

V = FunctionSpace(mesh, "P", 1)
psih = Function(V, name="psi_h").interpolate(psi)
uh = Function(V, name="u_h")
vh = TestFunction(V)
F = inner(grad(uh), grad(vh)) * dx  # f = 0
bcs = DirichletBC(V, Constant(0.0), "on_boundary")  # g = 0
problem = NonlinearVariationalProblem(F, uh, bcs)

sp = {"snes_type": "vinewtonrsls",  # <- explained next week
      "snes_converged_reason": None}  # 6 iterations for *this* grid
solver = NonlinearVariationalSolver(problem, solver_parameters=sp)
INF = Function(V).interpolate(Constant(PETSc.INFINITY))
solver.solve(bounds=(psih, INF))
VTKFile("cop.pvd").write(uh, psih)
```



FE for VIs: 2 new admissibility concerns

- the FE set-up for e.g. $X = H^1(\Omega)$:

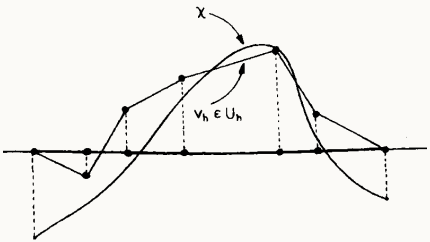
$$K = \{v : v \geq \psi\} \subset X$$
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- but $\psi_h = I_h\psi$ can be on either side of ψ, so

1. conformity is generally fragile, for any $X_h = P_k$:

$$K_h \not\subset K \text{ even when } X_h \subset X$$

   - Ciarlet's FE book includes a picture ↗



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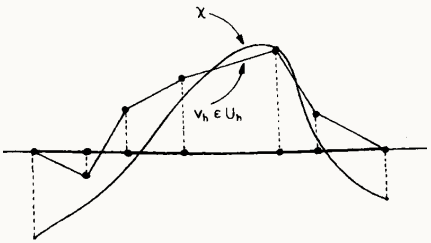
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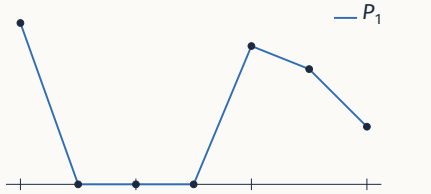
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$$v_{h,j} \leq w_{h,j} \implies v_h(x) \leq w_h(x)$$





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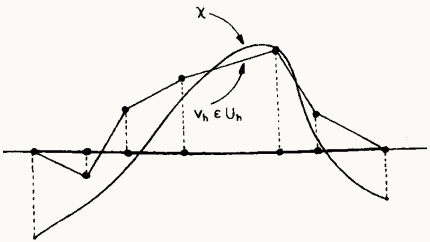
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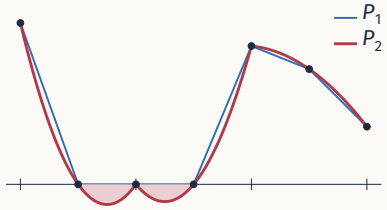
- P_1 enjoys nodal monotonicity:

$$v_{h,j} \leq w_{h,j} \implies v_h(x) \leq w_h(x)$$

2. which fails for P_k, $k \geq 2$ →

- e.g. $\psi = 0$: non-admissible $v_h \in P_2$ with $v_{h,j} \geq 0$





FE for VIs: Falk's lemma

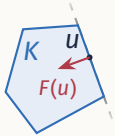
- for elliptic PDEs, the basic FE accuracy story starts with Cea's lemma
- what error norm bound applies to VIs?
  - the error is $u - u_h$

FE for VIs: Falk's lemma

- for elliptic PDEs, the basic FE accuracy story starts with Cea's lemma
- what error norm bound applies to VIs?
  - the error is $u - u_h$
- answer: Falk's generalization of Céa's lemma (next slide)
- stated using a standard set-up:
  - assume $K_h \subset X_h \subset X$ (but generally $K_h \not\subset K$)
  - suppose continuity for the bilinear form: $a(v,w) \le M\|v\|\|w\|$ for all $v, w \in X$
  - and coercivity: $a(v,v) \ge \alpha\|v\|^2$ for all $v \in X$
  - define a linear operator $A : X \to X'$ such that $a(v,w) = \langle Av, w\rangle$ for all $v, w \in X$
- for a source term $\ell \in X'$, the residual of $w \in X$ is $F(w) = Aw - \ell \in X'$

FE for VIs: Falk's lemma

- continuum problem for $u \in K$: $\quad a(u, v - u) \geq \langle \ell, v - u \rangle \quad \forall v \in K$
  - for the exact solution $u \in K$, generally $F(u) = Au - \ell \neq 0$
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Lemma (Falk, 1974)

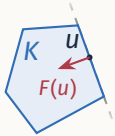
$$\|u - u_h\|_X^2 \leq \frac{M^2}{\alpha^2} \inf_{v_h \in K_h} \|u - v_h\|_X^2 + \frac{2}{\alpha} \left(\inf_{v \in K} \langle F(u), v - u_h \rangle + \inf_{v_h \in K_h} \langle F(u), v_h - u \rangle \right)$$

Corollary (Céa's lemma)

$K = X$: $\quad \|u - u_h\|_X \leq \frac{M}{\alpha} \inf_{v_h \in X_h} \|u - v_h\|_X$.

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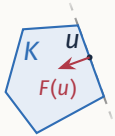
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Corollary (conforming K_h case)

$K_h \subset K$: *"$\inf_{v \in K}$" term can be replaced with zero*

FE for VIs: Falk's lemma

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Corollary (first-order convergence)

assuming quasi-uniform (regular) meshes: $\quad \|u - u_h\|_X \leq Ch$

summary

a variational inequality is:

- an inner-product description of inequality-constrained minimization
- a constrained generalization of a system of equations $F(u) = 0$, enabling existence
- a precise weak form for describing active-set and free-boundary problems
- solvable by finite elements (easily, in Firedrake)
- an interesting thing, worth learning about

thank you for listening!

questions?

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